

Evaluating Runtime Safety-Monitor Cascades: A Multi-Stack Measurement Pilot

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Abstract

Evaluation choices can materially change the apparent performance of a selective safety-monitor cascade. We study this effect using the existing three-level monitoring system as a development-only case study. The frozen pilot covers three monitor stacks and three data sources and compares four common evaluation choices: empirical false-positive screening versus a finite-sample risk bound, singleton versus dependency-aware grouping, original versus audited labels, and offline ranked acquisition versus a deployable threshold. We also replace the historical component-sum cost proxy with direct wall-clock end-to-end (E2E) latency measured at batch size one on a Tesla T4.

The FPR treatment changes 36 of 75 deployable policy conclusions, grouping changes 19 of 75, label choice changes 14 of 75, and ranked versus deployable routing changes 4 of 45 matched conclusions. Direct E2E timing is slightly higher than same-run component sums for every timed policy, although none of the 105 pairwise mean-cost orderings reverses. Two runtime route mismatches prevent a valid join between the frozen CPU recall/risk results and GPU cost measurements, so no Pareto claim is made. These results support an evaluation/measurement interpretation of the project rather than a router-superiority claim.

1 Introduction

Selective safety monitoring is attractive because a cheap monitor can handle routine cases while a stronger model is called only when needed. The difficulty is evaluation: a router can look effective when cost is approximated rather than measured, when the test ranking itself determines which examples are acquired, when near-duplicates are treated as independent, or when an empirical FPR is read as if it were a controlled risk.

The earlier v2 development screening in this project did not select a router, and several parts of that evidence require a narrower interpretation. Historical v2 costs were sums of measured component latencies rather than direct E2E policy latency. The grouping was effectively one group per example. The 35s quantity used in the bounded-cost analysis was not a mechanically enforced runtime limit, and the 1% cost-equivalence margin was an engineering choice rather than an externally justified confirmatory margin. Fresh calibration, joint FPR+cost certification, fresh source/time-separated confirmation, and multi-rater confirmation were not available. The protected legacy `final_test` and `held_out_shift` partitions remain unopened.

Rather than tuning the router again on the same data, we use the system as a case study in evaluation practice and ask whether more deployment-oriented evaluation choices change the apparent conclusion.

2 Pilot design

2.1 Data and monitor stacks

The authorized development view contains 1,687 prompt-response examples. The pilot uses three data sources: JailbreakBench judge-comparison, WildGuardTest, and XSTest-safe. For timing, a deterministic, label-blind sample was selected within each source and then closed over the frozen dependency groups, giving 363 examples in total: 120 from JailbreakBench, 123 from WildGuardTest, and 120 from XSTest-safe.

Three monitor stacks are evaluated:

1. rule $\rightarrow$ compact;
2. compact $\rightarrow$ Qwen3Guard;
3. rule + compact $\rightarrow$ Qwen3Guard.

The primary analysis uses audited harmful-response labels and the frozen dependency grouping. Five prespecified split seeds are used. Selective policies are evaluated at nominal acquisition targets of 0.10, 0.25, and 0.50. The offline ranked version is retained only as a diagnostic; deployment-oriented comparisons use a threshold that is fixed before the holdout is evaluated.

2.2 Risk treatment

A policy is not treated as acceptable merely because its observed holdout FPR is below 0.05. The primary pilot gate requires both the row-level one-sided 95% upper bound and the dependency-group false-positive event upper bound to be at most 0.05. This is still development-only evidence; it is not fresh external certification.

2.3 Timing protocol

Direct wall-clock E2E policy latency is the primary cost measurement. The same-run sum of monitor component timings is retained only so that the two cost definitions can be compared on exactly the same calls.

Timing was performed on one NVIDIA Tesla T4 at batch size one. All models were loaded before timing, 20 maximal-stack requests were used for warm-up and excluded from the measurements, CUDA stages were synchronized, and neither runtime capping nor post-hoc clipping was used. Fifteen deployable policies were measured on all 363 timing examples with a balanced rotating order, yielding 5,445 policy calls.

3 Results

3.1 The evaluation protocol changes the conclusion

Table 1 summarizes the main sensitivity analysis. The largest effect comes from FPR treatment. Forty of 75 deployable evaluations have empirical FPR at or below 0.05, but only four pass the frozen finite-sample row-and-dependency risk gate. This changes 36 policy-level pass/fail conclusions.

Table 1: Effect of each evaluation choice on the frozen pilot conclusions.

Evaluation comparison	Changed conclusions	Result
Empirical FPR vs finite-sample risk	36/75	Material
Singleton vs dependency grouping	19/75	Material
Original vs audited labels	14/75	Material
Ranked vs deployable routing	4/45	Material
Component sum vs direct E2E ordering	0/105	No ordering reversal

Dependency-aware grouping also matters. Replacing singleton grouping with the frozen near-duplicate grouping changes 19 of 75 conclusions. Using the audited labels instead of the original labels changes 14 of 75. Finally, the offline ranked diagnostic and the deployable threshold disagree on four of 45 matched risk conclusions.

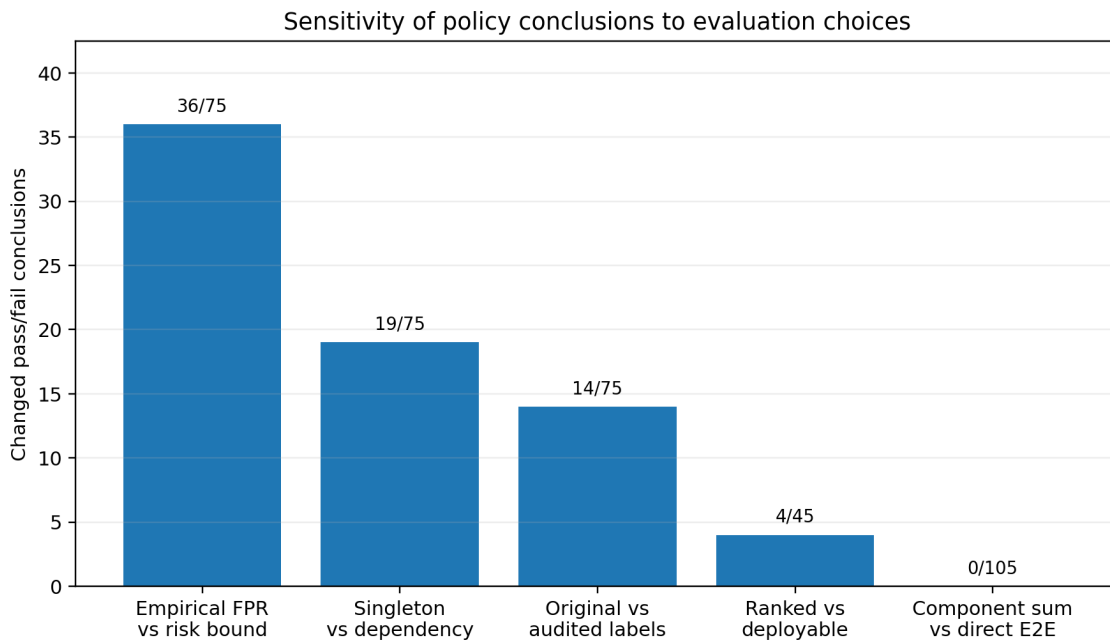


Figure 1: Sensitivity of the frozen pilot conclusions to the evaluation choice. The denominator is the number of matched evaluations for each comparison.

3.2 Ranked acquisition is not the same as a deployable threshold

The ranked diagnostic forces an exact acquisition count on the evaluation set. A reusable threshold does not. In all 45 matched ranked-versus-deployable comparisons, the realized holdout acquisition rate differs.

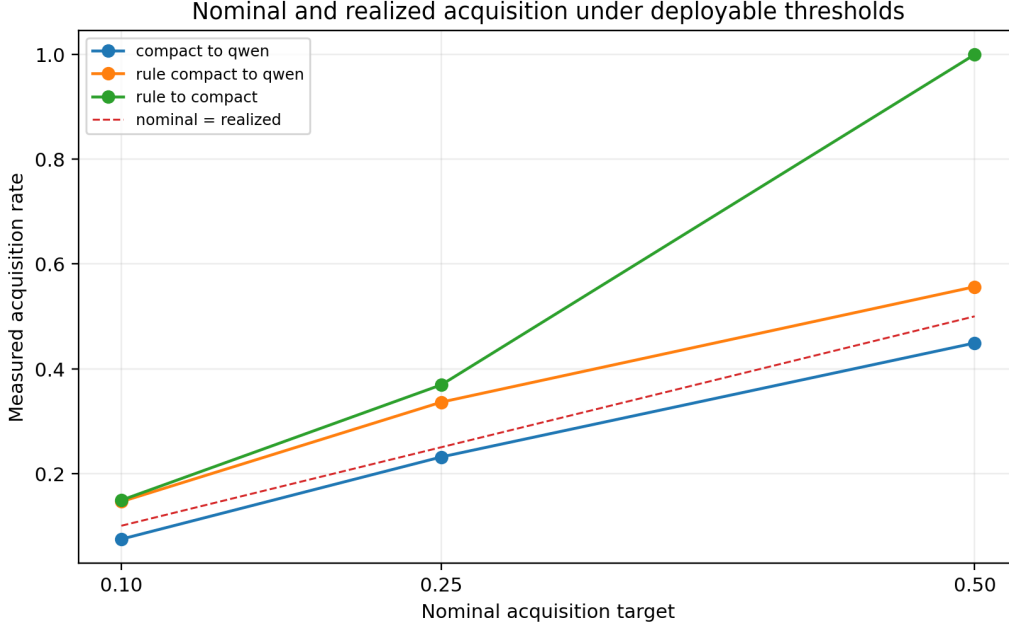


Figure 2: Nominal acquisition target and the acquisition rate realized by the frozen deployable threshold on the T4 timing sample. The rule-to-compact policy at the 0.50 target illustrates the effect of threshold ties: the measured acquisition rate becomes 1.0.

This distinction is methodological rather than a tuning problem. The ranked result answers what would happen if the evaluation set itself could be sorted. The threshold result answers what the frozen policy actually does when deployed on unseen examples.

3.3 Direct E2E cost and component sums

Direct E2E latency is slightly higher than the same-run component sum for every one of the 15 policies. The mean gap ranges from 0.028 to 0.308ms. However, this correction does not reverse the mean-cost ordering in the pilot: none of the 105 policy pairs changes order.

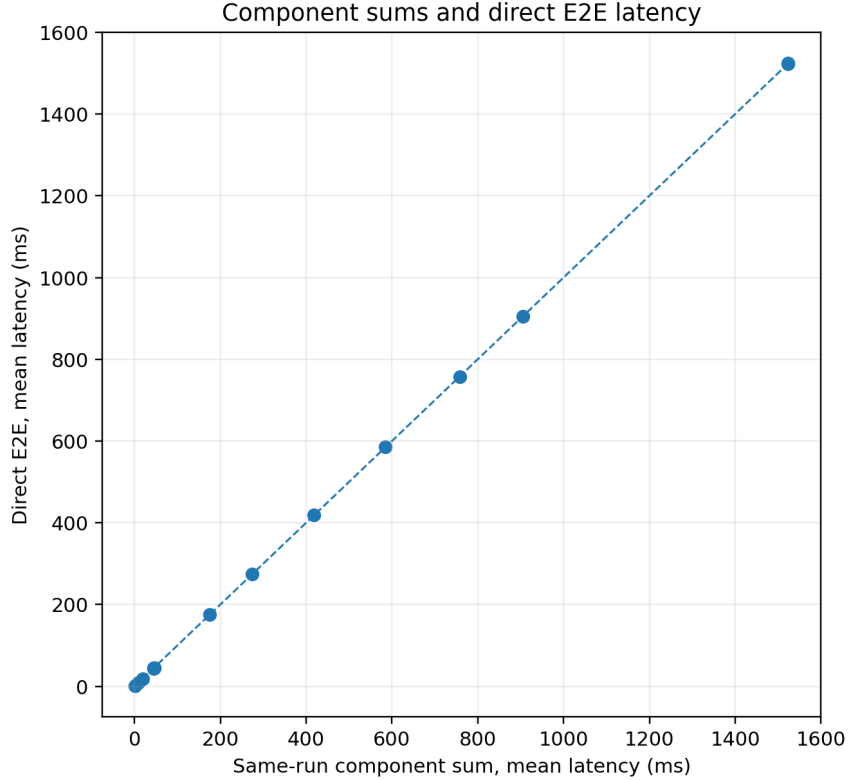


Figure 3: Direct wall-clock E2E latency against the same-run component sum. Points lie slightly above the identity line, showing small but consistent unattributed overhead.

The absence of a ranking reversal is a useful negative result. It does not make the component sum the correct estimand; it shows that, for this pilot, using the correct E2E measurement changes the absolute cost but not the ordering of the 15 policies.

3.4 Latency and tails

The Qwen-based stacks dominate runtime. Their full-information policies have mean direct E2E latency of about 1.52s, p95 around 2.22–2.23s, and maxima above 6.8s. The rule-to-compact full-information endpoint is roughly 44ms on average.

Table 2: Representative direct E2E timing results on Tesla T4.

Stack	Policy	Acquisition	Mean ms	p95 ms	p99 ms
compact to qwen	cheap only	0.000	45.8	80.1	81.4
compact to qwen	selective (0.25)	0.231	418.8	1814.5	2549.1
compact to qwen	full information	1.000	1523.2	2233.4	2654.5
rule compact to qwen	cheap only	0.000	45.2	80.1	81.7
rule compact to qwen	selective (0.25)	0.336	585.2	2099.2	2656.5
rule compact to qwen	full information	1.000	1524.0	2223.1	2644.6
rule to compact	cheap only	0.000	1.1	2.3	3.0
rule to compact	selective (0.25)	0.369	18.7	78.8	80.6
rule to compact	full information	1.000	44.0	76.9	80.4

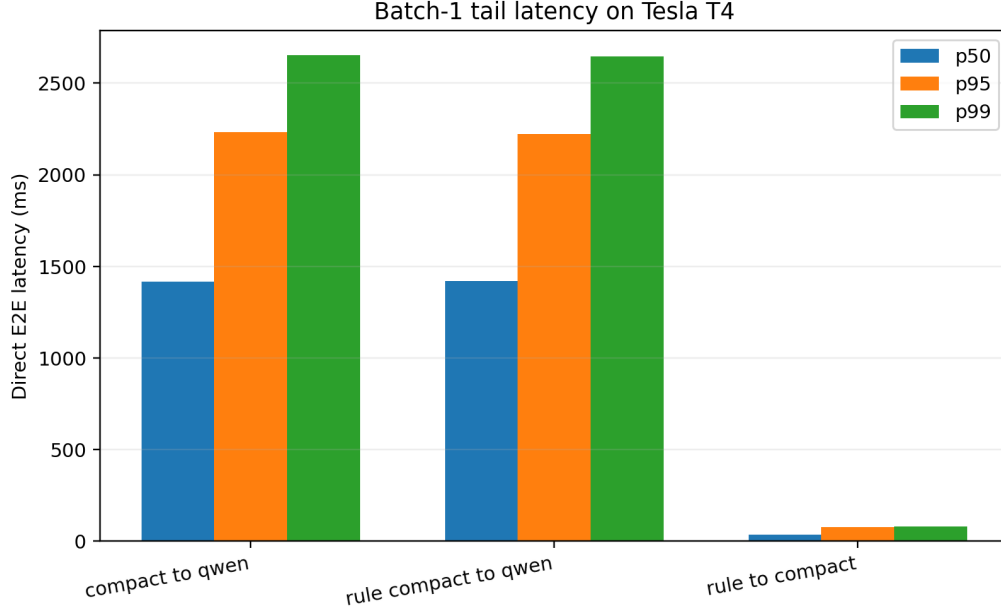


Figure 4: Median, p95, and p99 direct E2E latency for the three full-information policies on the Tesla T4.

Source-specific timing is reported separately because a pooled mean can hide substantial variation.

Table 3: Source-specific direct E2E latency for the full-information policies.

Stack	Source	n	Mean ms	p95 ms	p99 ms
Compact $\rightarrow$ Qwen	JailbreakBench	120	1561.5	2112.3	2177.1
Compact $\rightarrow$ Qwen	WildGuardTest	123	1741.3	2512.8	3028.7
Compact $\rightarrow$ Qwen	XSTest-safe	120	1261.2	1454.9	1606.0
Rule + compact $\rightarrow$ Qwen	JailbreakBench	120	1563.0	2124.1	2164.1
Rule + compact $\rightarrow$ Qwen	WildGuardTest	123	1742.4	2494.1	3028.7
Rule + compact $\rightarrow$ Qwen	XSTest-safe	120	1261.2	1450.0	1594.7
Rule $\rightarrow$ compact	JailbreakBench	120	45.4	77.3	80.3
Rule $\rightarrow$ compact	WildGuardTest	123	54.8	78.2	80.8
Rule $\rightarrow$ compact	XSTest-safe	120	31.6	41.8	42.4

3.5 Why no Pareto claim is reported

The T4 validation found two route mismatches and five prediction mismatches. The route mismatches are sufficient to invalidate the frozen join between the CPU recall/risk evidence and the GPU cost evidence: `cpu_cost_join_valid=false`. We therefore do not combine those two evidence streams to claim Pareto dominance.

The thresholds are therefore left unchanged, and the mismatched run is reported as observed rather than repeated or adjusted after the fact.

4 Discussion

The pilot shows that several ordinary evaluation choices are large enough to change the apparent result of a safety-monitor routing study. In this case, the strongest changes come from finite-sample risk treatment, dependence-aware grouping, and label audit. Ranked acquisition also gives a different operational picture from a reusable threshold. By contrast, direct E2E measurement does not reverse the mean-cost ordering in this particular pilot, even though it remains the correct latency estimand.

The historical v2 result should be read narrowly. No signed-value family had a jointly feasible development operating point, and no router was selected. The component-cost tables remain useful as historical measurements, but they are not direct-E2E deployment costs. Likewise, the 35 s normalization and the 1% equivalence margin should not be presented as a certified runtime bound or a generally justified confirmatory margin.

The pilot itself is also deliberately limited. It does not provide fresh joint FPR+cost certification, fresh source/time-separated confirmation, or multi-rater confirmation. The observed finite-sample bounds are internal development safeguards, not external certificates. The work also does not establish arbitrary distribution-shift robustness, correlated-failure detection, or production readiness.

5 Conclusion

Using the existing monitor cascade as a case study, the pilot finds that evaluation methodology can materially change which policies appear acceptable. Empirical FPR screening, singleton grouping, original labels, and offline ranking each produce conclusions that differ from their more deployment-oriented counterparts. Direct E2E timing adds a small positive cost relative to component sums but does not change the mean-cost ordering in this run.

The appropriate conclusion at this stage is therefore about measurement, not router superiority. The pilot provides a reproducible example of how safety-monitor cascade results depend on the evaluation protocol and why cost, dependence, risk, labels, and deployability need to be reported explicitly.

A Full T4 timing results

For completeness, the fifteen deployable policies are listed below by stack. These supplementary tables provide the full timing record behind the representative results in the main text and do not introduce additional claims.

Table 4: Rule $\rightarrow$ compact.

Policy	Acquisition	Mean ms	p95 ms	p99 ms	Route match
cheap only	0.000	1.1	2.3	3.0	1.0000
selective (0.10)	0.149	8.5	72.5	81.5	1.0000
selective (0.25)	0.369	18.7	78.8	80.6	1.0000
selective (0.50)	1.000	45.7	79.9	81.5	1.0000
full information	1.000	44.0	76.9	80.4	1.0000

Table 5: Compact $\rightarrow$ Qwen.

Policy	Acquisition	Mean ms	p95 ms	p99 ms	Route match
cheap only	0.000	45.8	80.1	81.4	1.0000
selective (0.10)	0.074	175.3	1375.9	2476.7	1.0000
selective (0.25)	0.231	418.8	1814.5	2549.1	0.9972
selective (0.50)	0.449	757.8	2193.0	2629.5	1.0000
full information	1.000	1523.2	2233.4	2654.5	1.0000

Table 6: Rule + compact $\rightarrow$ Qwen.

Policy	Acquisition	Mean ms	p95 ms	p99 ms	Route match
cheap only	0.000	45.2	80.1	81.7	1.0000
selective (0.10)	0.146	274.0	1682.8	2311.8	1.0000
selective (0.25)	0.336	585.2	2099.2	2656.5	1.0000
selective (0.50)	0.556	904.8	2153.1	2653.7	0.9972
full information	1.000	1524.0	2223.1	2644.6	1.0000